DESCENT OF THE FLAT-AT-PRIMES CONDITION ALONG A SURJECTIVE MAP ON THE RELEVANT CLOSED SETS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness at a prime of the source can be detected after a flat base change to a prime of the target. Algebra, Lemma \ref{algebra-lemma-base-change-flat-up-down}, cited at proof lines 4287-4289. Carries flatness from the localised base change back to the localisation of the source.
- The induced map on closed sets is surjective, so every prime in the relevant locus of the source has a prime of the target above it. assumption (1) of the lemma, used at proof lines 4275-4277. Produces the prime of the target and the flat local map needed for the chase.

Source reference. Original lemma 05LS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7246–7246. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05LP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7243–7243. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05LQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7244–7244. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Source reference. Original lemma 00MQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 832–832. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *In Situation 05LP let $R' \to R''$ be an R-algebra map. Let $I' \subset R'$ and $I'R'' \subset I'' \subset R''$ be ideals. Assume*

(1) *the map $V(I'') \to V(I')$ induced by $\mathrm{Spec}(R'') \to \mathrm{Spec}(R')$ is surjective, and*
(2) *$R''_{\mathfrak{p}''}$ is flat over R' for all primes $\mathfrak{p}'' \in V(I'')$.*

If (05LQ) holds for (R'', I''), then (05LQ) holds for (R', I').

Proof. Assume (05LQ) holds for (R'', I''). Pick a prime $I'S' + JS' \subset \mathfrak{q}' \subset S'$. Let $I' \subset \mathfrak{p}' \subset R'$ be the corresponding prime of R'. By assumption there exists a prime $\mathfrak{p}'' \in V(I'')$ of R'' lying over $\mathfrak{p}'$ and $R'_{\mathfrak{p}'} \to R''_{\mathfrak{p}''}$ is flat. Choose a prime $\overline{\mathfrak{q}}'' \subset \kappa(\mathfrak{q}') \otimes_{\kappa(\mathfrak{p}')} \kappa(\mathfrak{p}'')$. This corresponds to a prime $\mathfrak{q}'' \subset S'' = S' \otimes_{R'} R''$ which lies over $\mathfrak{q}'$ and over $\mathfrak{p}''$. In particular we see that $I''S'' \subset \mathfrak{q}''$ and that $JS'' \subset \mathfrak{q}''$. Note that $(S' \otimes_{R'} R'')_{\mathfrak{q}''}$ is a localization of $S'_{\mathfrak{q}'} \otimes_{R'_{\mathfrak{p}'}} R''_{\mathfrak{p}''}$. By assumption the module $(M' \otimes_{R'} R'')_{\mathfrak{q}''}$ is flat over $R''_{\mathfrak{p}''}$. Hence Algebra, Lemma 00MQ implies that $M'_{\mathfrak{q}'}$ is flat over $R'_{\mathfrak{p}'}$ which is what we wanted to prove. □